## $L_1$ Problem: Maximize number of contacts

$$\max\ \mathbf{1}^\mathsf{T}\boldsymbol{x}$$

$$\text{s.t. } A\boldsymbol{x} \le \boldsymbol{b}, \qquad \boldsymbol{x} \in \{0,1\}^N$$

## $L_2$ Problem: Minimize Square-sum Revisit Time

$$\min\ \mathbf{1}^\mathsf{T} G^2 \boldsymbol{z}$$

$$\text{s.t. } A\boldsymbol{x} \le \boldsymbol{b}, \qquad \boldsymbol{x} \in \{0,1\}^N$$

$$\boldsymbol{z} \ge E\boldsymbol{x} + \boldsymbol{f} - \mathbf{1}, \qquad \boldsymbol{z} \ge \mathbf{0}$$

$$\boldsymbol{z} \in R^M, M = \sum_{j=1}^{q} \frac{(|G_j|+2)(|G_j|+1)}{2}$$

## $L_\infty$ Problem: Minimize Maximum Revisit Time

$$\min\ R$$

$$\text{s.t. } A\boldsymbol{x} \le \boldsymbol{b}, \qquad \boldsymbol{x} \in \{0,1\}^N$$

$$R\mathbf{1} \ge G(E\boldsymbol{x} + \boldsymbol{f} - \mathbf{1})$$

## Constants

$$A = \begin{bmatrix} E^2_{S_1,x} E^1_{S_1} \\ E^2_{S_2,x} E^1_{S_2} \\ \vdots \\ E^2_{S_p,x} E^1_{S_p} \end{bmatrix}, \qquad \boldsymbol{b} = \mathbf{1} + \frac{1}{\tau} \begin{bmatrix} E^2_{S_1,t} E^1_{S_1} \\ E^2_{S_2,t} E^1_{S_2} \\ \vdots \\ E^2_{S_p,t} E^1_{S_p} \end{bmatrix} \boldsymbol{t}$$

$$E = \begin{bmatrix} E^2_{G_1,x} E^1_{G_1} \\ E^2_{G_2,x} E^1_{G_2} \\ \vdots \\ E^2_{G_q,x} E^1_{G_q} \end{bmatrix} \begin{bmatrix} \mathbf{0}_{1\times N} \\ \mathbf{I}_{N\times N} \\ \mathbf{0}_{1\times N} \end{bmatrix}, \quad \boldsymbol{f} = \begin{bmatrix} E^2_{G_1,x} E^1_{G_1} \\ E^2_{G_2,x} E^1_{G_2} \\ \vdots \\ E^2_{G_q,x} E^1_{G_q} \end{bmatrix} \begin{bmatrix} 1 \\ \mathbf{0}_{N\times 1} \\ 1 \end{bmatrix}$$

$$G = \operatorname{diag}\left( \begin{bmatrix} E^2_{G_1,t} E^1_{G_1} \\ E^2_{G_2,t} E^1_{G_2} \\ \vdots \\ E^2_{G_q,t} E^1_{G_q} \end{bmatrix} \begin{bmatrix} t_{\text{start}} \\ \boldsymbol{t} \\ t_{\text{end}} \end{bmatrix} \right)$$